

# A Factorially Weighted Constant-Term Theorem on Algebraic Tori and the Gaussian Moments Conjecture in Dimension Two

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## Abstract

For an integer  $r \geq 1$ , let

$$\mathcal{A}_r = \mathbb{C}[u, z_1^{\pm 1}, \dots, z_r^{\pm 1}]$$

and define the factorially weighted constant-term functional by

$$\Gamma_r(u^n z^\mu) = n! \mathbf{1}_{\mu=0} \quad (n \geq 0, \mu \in \mathbb{Z}^r).$$

For  $F \in \mathcal{A}_r$ , let  $\Omega(F) \subset \mathbb{Z}^r$  be its torus-weight support. We prove that the following are equivalent:  $\Gamma_r(F^m) = 0$  for all sufficiently large  $m$ ;  $\Gamma_r(F^m) = 0$  for every  $m \geq 1$ ; and  $0 \notin \text{conv } \Omega(F)$ . Under these conditions,  $\Gamma_r(GF^m) = 0$  for every fixed  $G \in \mathcal{A}_r$  and all sufficiently large  $m$ . Hence  $\ker \Gamma_r$  is a Mathieu–Zhao subspace for every torus rank. More quantitatively, if  $0 \in \text{conv } \Omega(F)$ , then there is an integer  $k \geq 1$  such that  $\Gamma_r(F^{kp}) \neq 0$  for every sufficiently large prime  $p$ . The proof combines the multivariable constant-term theorem of Duistermaat and van der Kallen, an exposed-face construction, and a reduction-modulo- $p$  argument that isolates the lowest balanced coefficient through factorial weights. As a specialization, an explicit embedding of the two-variable Gaussian functional into  $\Gamma_1$  proves the Gaussian Moments Conjecture in dimension two and the corresponding orthogonal nullcone classification.

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## 1 Introduction

For  $r \geq 1$ , consider the Laurent polynomial algebra with one distinguished nonnegative radial variable

$$\mathcal{A}_r = \mathbb{C}[u, z_1^{\pm 1}, \dots, z_r^{\pm 1}].$$

Writing  $z^\mu = z_1^{\mu_1} \cdots z_r^{\mu_r}$  for  $\mu = (\mu_1, \dots, \mu_r) \in \mathbb{Z}^r$ , define a linear functional

$$\Gamma_r : \mathcal{A}_r \longrightarrow \mathbb{C}, \quad \Gamma_r(u^n z^\mu) = n! \mathbf{1}_{\mu=0}.$$

Equivalently, with normalized Haar measure on  $\mathbb{T}^r$ ,

$$\Gamma_r(F) = \int_0^\infty \int_{\mathbb{T}^r} F(u, z) e^{-u} dz du. \quad (1.1)$$

Thus  $\Gamma_r$  first extracts torus weight zero and then applies the Gamma moment  $u^n \mapsto n!$ .

For

$$F = \sum_{\substack{n \geq 0 \\ \mu \in \mathbb{Z}^r}} c_{n,\mu} u^n z^\mu \in \mathcal{A}_r,$$

define its torus-weight support by

$$\Omega(F) = \{\mu \in \mathbb{Z}^r : c_{n,\mu} \neq 0 \text{ for some } n \geq 0\}. \quad (1.2)$$

We use the convention  $\Omega(0) = \emptyset$  and  $\text{conv } \emptyset = \emptyset$ .

The main theorem classifies all polynomials whose sufficiently high power moments vanish.

**Theorem 1.1** (Factorially weighted torus theorem). *Let  $r \geq 1$  and  $F \in \mathcal{A}_r$ . The following conditions are equivalent.*

- (i)  $\Gamma_r(F^m) = 0$  for all sufficiently large integers  $m$ ;
- (ii)  $\Gamma_r(F^m) = 0$  for every integer  $m \geq 1$ ;
- (iii)  $0 \notin \text{conv } \Omega(F)$ .

Moreover, whenever these conditions hold, for every  $G \in \mathcal{A}_r$  one has

$$\Gamma_r(GF^m) = 0 \quad (m \gg 0).$$

The quantitative form is stronger than the failure of eventual vanishing.

**Theorem 1.2** (Prime-index nonvanishing). *Let  $r \geq 1$  and let  $F \in \mathcal{A}_r$  be nonzero. If  $0 \in \text{conv } \Omega(F)$ , then there exist integers  $k \geq 1$  and  $p_0 \geq 1$  such that, for every rational prime  $p > p_0$ ,*

$$\Gamma_r(F^{kp}) \neq 0.$$

Recall that a linear subspace  $M$  of a commutative  $\mathbb{C}$ -algebra  $B$  is a *Mathieu–Zhao subspace* if, whenever  $a^m \in M$  for every  $m \geq 1$ , one has  $ba^m \in M$  for all sufficiently large  $m$ , for every  $b \in B$ ; see [5]. The mixed-moment assertion in Theorem 1.1 immediately gives the following consequence.

**Corollary 1.3.** *For every  $r \geq 1$ , the kernel of  $\Gamma_r$  is a Mathieu–Zhao subspace of  $\mathcal{A}_r$ .*

The proof has two independent components. First, if the origin lies in the convex hull of the torus weights, a linear-programming argument exposes the lowest face capable of producing total torus weight zero. The theorem of Duistermaat and van der Kallen then supplies a nonzero balanced coefficient on a power of the face polynomial. Second, after normalizing by the factorial attached to this lowest coefficient, reduction modulo a prime shows that all higher radial terms vanish while the distinguished term survives. The exposed-face argument works in arbitrary torus rank; the arithmetic isolation remains one-dimensional because there is only one radial exponent.

The two-dimensional Gaussian theorem is obtained from the rank-one case. If  $X, Y$  are independent standard real Gaussian random variables, set

$$Z = \frac{X + iY}{\sqrt{2}}, \quad W = \frac{X - iY}{\sqrt{2}}. \quad (1.3)$$

The Gaussian contraction rule is

$$\mathbb{E}[Z^a W^b] = a! \mathbf{1}_{a=b}.$$

The injective algebra homomorphism

$$\iota : \mathbb{C}[Z, W] \hookrightarrow \mathcal{A}_1, \quad \iota(Z) = z, \quad \iota(W) = uz^{-1}, \quad (1.4)$$

intertwines Gaussian expectation with  $\Gamma_1$ . Consequently, Theorem 1.1 yields a complete classification of Gaussian moment-null polynomials in two variables, proves GMC(2), and verifies the corresponding nullcone conjecture of Derksen, van den Essen, and Zhao.

## 2 The Gamma functional and torus weights

For the remainder of the paper, fix  $r \geq 1$ . Every element of  $\mathcal{A}_r$  has a unique finite expansion

$$F = \sum_{\substack{n \geq 0 \\ \mu \in \mathbb{Z}^r}} c_{n,\mu} u^n z^\mu.$$

The functional is therefore

$$\Gamma_r(F) = \sum_{n \geq 0} n! [u^n z^0] F. \quad (2.1)$$

The integral representation (1.1) follows from

$$\int_{\mathbb{T}^r} z^\mu dz = \mathbf{1}_{\mu=0}, \quad \int_0^\infty u^n e^{-u} du = n!.$$

The torus weights give a  $\mathbb{Z}^r$ -grading

$$\mathcal{A}_r = \bigoplus_{\mu \in \mathbb{Z}^r} (\mathcal{A}_r)_\mu, \quad (\mathcal{A}_r)_\mu = \text{span}_{\mathbb{C}}\{u^n z^\mu : n \geq 0\}. \quad (2.2)$$

By definition,  $\Gamma_r$  vanishes on every nonzero weight space.

**Lemma 2.1** (Strict separation and mixed moments). *Suppose that  $F \in \mathcal{A}_r$  satisfies  $0 \notin \text{conv } \Omega(F)$ . Then*

$$\Gamma_r(F^m) = 0 \quad (m \geq 1),$$

and, for every  $G \in \mathcal{A}_r$ ,

$$\Gamma_r(GF^m) = 0 \quad (m \gg 0).$$

*Proof.* The assertion is immediate for  $F = 0$ , so assume  $F \neq 0$ . Since  $\text{conv } \Omega(F)$  is compact and does not contain the origin, strict separation gives a linear functional  $\lambda : \mathbb{R}^r \rightarrow \mathbb{R}$  and a number  $\delta > 0$  such that

$$\lambda(\mu) \geq \delta \quad (\mu \in \Omega(F)).$$

Every torus weight occurring in  $F^m$  therefore has  $\lambda$ -value at least  $m\delta$  and is nonzero. Hence  $F^m$  has no weight-zero component and  $\Gamma_r(F^m) = 0$ .

If  $G \neq 0$ , let

$$c = \min\{\lambda(\eta) : \eta \in \Omega(G)\}.$$

Every weight in  $GF^m$  has  $\lambda$ -value at least  $c + m\delta$ , which is positive for all sufficiently large  $m$ . Thus  $GF^m$  has no weight-zero component and  $\Gamma_r(GF^m) = 0$ . The case  $G = 0$  is trivial.  $\square$

### 3 The lowest balanced exposed face

For a Laurent polynomial  $f \in \mathbb{C}[z_1^{\pm 1}, \dots, z_r^{\pm 1}]$ , write  $\text{CT}(f)$  for its constant term. We use the full torus form of the theorem of Duistermaat and van der Kallen.

**Theorem 3.1** (Duistermaat–van der Kallen). *Let  $f \in \mathbb{C}[z_1^{\pm 1}, \dots, z_r^{\pm 1}]$ . If*

$$\text{CT}(f^m) = 0 \quad \text{for every } m \geq 1,$$

*then*

$$0 \notin \text{conv supp}(f).$$

*Equivalently, if  $0 \in \text{conv supp}(f)$ , then  $\text{CT}(f^k) \neq 0$  for at least one integer  $k \geq 1$ .*

*Reference.* This is [2, Theorem 1]. □

We next record the supporting-face statement used to expose the least radial degree compatible with total torus weight zero.

**Lemma 3.2** (Lowest balanced face). *Let  $S$  be a finite set, let  $w : S \rightarrow \mathbb{R}^r$ , and let  $n : S \rightarrow \mathbb{R}$ . Assume that  $0 \in \text{conv } w(S)$ . Define*

$$\rho = \min \left\{ \sum_{s \in S} t_s n(s) : t_s \geq 0, \sum_{s \in S} t_s = 1, \sum_{s \in S} t_s w(s) = 0 \right\}. \quad (3.1)$$

*Then there exists  $\nu \in \mathbb{R}^r$  such that*

$$n(s) + \langle \nu, w(s) \rangle \geq \rho \quad (s \in S). \quad (3.2)$$

*If*

$$S_0 = \{s \in S : n(s) + \langle \nu, w(s) \rangle = \rho\},$$

*then  $0 \in \text{conv } w(S_0)$ .*

*Proof.* The optimization problem (3.1) is a feasible bounded linear program. Its dual is

$$\max \{ \rho' : \rho' \leq n(s) + \langle \nu, w(s) \rangle \text{ for every } s \in S \},$$

with variables  $(\rho', \nu) \in \mathbb{R} \times \mathbb{R}^r$ . Strong linear-programming duality gives an optimal pair  $(\rho, \nu)$ ; see, for example, [3, Chapter 7]. If  $(t_s)$  is an optimal solution of (3.1), then

$$0 = \sum_{s \in S} t_s (n(s) + \langle \nu, w(s) \rangle - \rho).$$

Every summand is nonnegative, so  $t_s > 0$  only for  $s \in S_0$ . Since the  $t_s$  sum to one and their  $w$ -weighted average is zero, it follows that  $0 \in \text{conv } w(S_0)$ . □

**Proposition 3.3** (Balanced face extraction). *Let  $F \in \mathcal{A}_r$  be nonzero and assume  $0 \in \text{conv } \Omega(F)$ . Then there exist*

- *an integer  $k \geq 1$ ,*
- *an integer  $d \geq 0$ ,*
- *a vector  $\nu \in \mathbb{R}^r$ , and*

• a nonzero complex number  $C$ ,  
such that, for  $R = F^k$ ,

(a) every monomial  $u^N z^\mu$  occurring in  $R$  satisfies

$$N + \langle \nu, \mu \rangle \geq d; \quad (3.3)$$

(b)

$$[u^d z^0]R = C \neq 0. \quad (3.4)$$

*Proof.* Write

$$F = \sum_{(n, \mu) \in S} c_{n, \mu} u^n z^\mu, \quad S = \text{supp}(F) \subset \mathbb{Z}_{\geq 0} \times \mathbb{Z}^r.$$

Apply Lemma 3.2 with

$$w(n, \mu) = \mu, \quad n(n, \mu) = n.$$

Thus there are  $\rho \in \mathbb{R}$  and  $\nu \in \mathbb{R}^r$  such that

$$\ell(n, \mu) := n + \langle \nu, \mu \rangle \geq \rho \quad ((n, \mu) \in S), \quad (3.5)$$

and the equality set

$$S_0 = \{(n, \mu) \in S : \ell(n, \mu) = \rho\}$$

satisfies  $0 \in \text{conv}\{\mu : (n, \mu) \in S_0\}$ .

Projection from  $S_0$  to  $\mathbb{Z}^r$  is injective. Indeed, if  $(n, \mu)$  and  $(n', \mu)$  both lie on the face, then

$$n + \langle \nu, \mu \rangle = n' + \langle \nu, \mu \rangle = \rho,$$

so  $n = n'$ . Consequently, for the face polynomial

$$H(u, z) = \sum_{(n, \mu) \in S_0} c_{n, \mu} u^n z^\mu$$

and the Laurent polynomial

$$h(z) = \sum_{(n, \mu) \in S_0} c_{n, \mu} z^\mu, \quad (3.6)$$

there is no cancellation when equal Laurent monomials are collected, and

$$0 \in \text{conv supp}(h).$$

By the contrapositive of Theorem 3.1, there is an integer  $k \geq 1$  such that

$$C := \text{CT}(h^k) \neq 0. \quad (3.7)$$

Every contribution to the constant term in (3.7) comes from a product of  $k$  face monomials whose total torus weight is zero. Because every factor lies on  $\ell = \rho$ , additivity gives

$$\ell(N, 0) = N = k\rho.$$

Thus every balanced product of  $k$  face monomials has the same radial degree

$$d := k\rho \in \mathbb{Z}_{\geq 0},$$

and therefore

$$[u^d z^0]H^k = \text{CT}(h^k) = C \neq 0. \quad (3.8)$$

Set  $R = F^k$ . Every monomial of  $R$  has  $\ell$ -degree at least  $k\rho = d$ . A product in the expansion of  $F^k$  that uses at least one monomial outside  $S_0$  has  $\ell$ -degree strictly greater than  $d$  and hence cannot contribute to  $u^d z^0$ . Equation (3.8) therefore gives  $[u^d z^0]R = C$ , and (3.3) follows from additivity.  $\square$

## 4 Factorial isolation by reduction modulo a prime

We first record a spreading-out fact in the form required below.

**Lemma 4.1** (Good reductions). *Let  $B$  be a finitely generated integral domain over  $\mathbb{Z}$  of characteristic zero. Then there is an integer  $N \geq 1$  such that, for every prime  $p \nmid N$ , there exist a field  $K_p$  of characteristic  $p$  and a unital ring homomorphism*

$$\varphi_p : B \longrightarrow K_p.$$

*In particular, every unit of  $B$  has nonzero image in  $K_p$ .*

*Proof.* The morphism

$$\pi : \operatorname{Spec}(B) \longrightarrow \operatorname{Spec}(\mathbb{Z})$$

is dominant because the zero ideal of the domain  $B$  contracts to the zero ideal of  $\mathbb{Z}$ . Since  $\mathbb{Z}$  is Noetherian, the finite-type  $\mathbb{Z}$ -algebra  $B$  is finitely presented. Chevalley's theorem shows that the image of  $\pi$  is constructible; see [4, Tag 00FE].

This image contains the generic point of the irreducible Noetherian space  $\operatorname{Spec}(\mathbb{Z})$ . A constructible subset of an irreducible Noetherian space that contains the generic point contains a nonempty open subset. It follows that the image of  $\pi$  contains  $D(N)$  for some integer  $N \geq 1$ .

For every prime  $p \nmid N$ , choose a prime ideal  $\mathfrak{q} \subset B$  with  $\mathfrak{q} \cap \mathbb{Z} = (p)$  and set

$$K_p = \operatorname{Frac}(B/\mathfrak{q}).$$

Then  $K_p$  has characteristic  $p$ , and the composite  $B \rightarrow B/\mathfrak{q} \hookrightarrow K_p$  is the required homomorphism. A unit cannot have zero image under a unital homomorphism to a field.  $\square$

The next lemma is the arithmetic core of the argument.

**Lemma 4.2** (Factorial isolation). *Let*

$$R = \sum_{\substack{n \geq 0 \\ \mu \in \mathbb{Z}^r}} r_{n,\mu} u^n z^\mu \in \mathcal{A}_r.$$

*Suppose that there are  $\nu \in \mathbb{R}^r$  and an integer  $d \geq 0$  such that*

$$n + \langle \nu, \mu \rangle \geq d \quad \text{whenever } r_{n,\mu} \neq 0, \tag{4.1}$$

*and suppose*

$$C := r_{d,0} \neq 0. \tag{4.2}$$

*Then*

$$\Gamma_r(R^p) \neq 0$$

*for every sufficiently large prime  $p$ .*

*Proof.* Let

$$B = \mathbb{Z}[\{r_{n,\mu}\}, C^{-1}] \subset \mathbb{C}, \tag{4.3}$$

where only the finitely many nonzero coefficients of  $R$  are included. This is a finitely generated integral domain of characteristic zero. Choose  $N$  as in Lemma 4.1.

Fix a prime  $p \nmid N$ . Every torus-weight-zero monomial  $u^j z^0$  occurring in  $R^p$  satisfies

$$j = j + \langle \nu, 0 \rangle \geq pd,$$

because the degree  $(n, \mu) \mapsto n + \langle \nu, \mu \rangle$  is additive and all monomials of  $R$  have degree at least  $d$ . Define

$$G_p = \sum_{j \geq pd} [u^j z^0] R^p \frac{j!}{(pd)!} \in B. \quad (4.4)$$

By (2.1),

$$\Gamma_r(R^p) = (pd)! G_p. \quad (4.5)$$

Let  $\varphi_p : B \rightarrow K_p$  be supplied by Lemma 4.1, and place a bar over reductions to  $K_p$ . In characteristic  $p$ , the Frobenius identity gives

$$\overline{R^p} = \bar{R}^p = \sum_{n, \mu} \bar{r}_{n, \mu}^p u^{pn} z^{p\mu}. \quad (4.6)$$

Only terms with  $\mu = 0$  can have torus weight zero after reduction, and therefore

$$\overline{G_p} = \sum_{n \geq d} \bar{r}_{n, 0}^p \frac{(pn)!}{(pd)!} \quad \text{in } K_p. \quad (4.7)$$

For  $n = d$ , the factorial ratio is 1, and the corresponding term is  $\bar{C}^p$ . If  $n > d$ , then

$$\frac{(pn)!}{(pd)!} = (pd+1)(pd+2) \cdots (pn)$$

is divisible by  $p$ , since the displayed product contains the factor  $p(d+1)$ . Hence all terms with  $n > d$  vanish in  $K_p$ , while

$$\overline{G_p} = \bar{C}^p \neq 0. \quad (4.8)$$

The last inequality holds because  $C$  is a unit in  $B$ . Thus  $G_p \neq 0$ , and (4.5) implies  $\Gamma_r(R^p) \neq 0$ . This holds for every prime  $p \nmid N$ , hence for every sufficiently large prime.  $\square$

*Remark 4.3.* The normalization by  $(pd)!$  is essential: it converts the higher radial terms into factorial ratios divisible by  $p$ , while the distinguished lowest balanced coefficient retains ratio 1.

## 5 Proof of the main theorem

*Proof of Theorem 1.2.* Apply Proposition 3.3 to obtain  $R = F^k$ ,  $d$ ,  $\nu$ , and  $C \neq 0$  satisfying (3.3) and (3.4). Lemma 4.2 gives

$$\Gamma_r(F^{kp}) = \Gamma_r(R^p) \neq 0$$

for every sufficiently large prime  $p$ .  $\square$

*Proof of Theorem 1.1.* The implication (ii) $\Rightarrow$ (i) is immediate. Lemma 2.1 proves (iii) $\Rightarrow$ (ii) and also proves the mixed-moment conclusion.

It remains to prove (i) $\Rightarrow$ (iii). The case  $F = 0$  is immediate. Suppose  $F \neq 0$  and  $\Gamma_r(F^m) = 0$  for all sufficiently large  $m$ . If  $0 \in \text{conv } \Omega(F)$ , Theorem 1.2 gives an integer  $k \geq 1$  such that

$$\Gamma_r(F^{kp}) \neq 0$$

for every sufficiently large prime  $p$ , contradicting eventual vanishing. Therefore  $0 \notin \text{conv } \Omega(F)$ .  $\square$

*Proof of Corollary 1.3.* Suppose  $F^m \in \ker \Gamma_r$  for every  $m \geq 1$ . By Theorem 1.1,  $0 \notin \text{conv } \Omega(F)$ . The mixed-moment part of the theorem gives  $\Gamma_r(GF^m) = 0$  for all sufficiently large  $m$  and every  $G \in \mathcal{A}_r$ . This is precisely the Mathieu–Zhao property.  $\square$

Theorem 1.1 also identifies the radical of the kernel in support terms.

**Corollary 5.1** (Radical classification). *For every  $r \geq 1$ ,*

$$\{F \in \mathcal{A}_r : \Gamma_r(F^m) = 0 \text{ for } m \gg 0\} = \{F \in \mathcal{A}_r : 0 \notin \text{conv } \Omega(F)\}.$$

*If  $F \neq 0$  belongs to this set, there is a linear functional  $\lambda : \mathbb{R}^r \rightarrow \mathbb{R}$  that is strictly positive on every weight of  $F$ .*

*Proof.* The equality is Theorem 1.1. The final assertion is the strict separation theorem applied to the compact polytope  $\text{conv } \Omega(F)$  and the origin.  $\square$

## 6 The two-dimensional Gaussian specialization

Let  $X, Y$  be independent standard real Gaussian random variables and define  $Z, W$  by (1.3). Since the change of variables is invertible over  $\mathbb{C}$ , we identify  $\mathbb{C}[X, Y]$  with  $\mathbb{C}[Z, W]$ .

**Lemma 6.1** (Gaussian contraction rule). *For all  $a, b \in \mathbb{Z}_{\geq 0}$ ,*

$$\mathbb{E}[Z^a W^b] = \begin{cases} a!, & a = b, \\ 0, & a \neq b. \end{cases} \quad (6.1)$$

*Proof.* Write  $X = r \cos \theta$  and  $Y = r \sin \theta$ . Under standard Gaussian measure,  $\theta$  is uniform on  $[0, 2\pi)$  and independent of  $U = r^2/2$ , while  $U$  has density  $e^{-u} du$  on  $[0, \infty)$ . Moreover,

$$Z = \sqrt{U} e^{i\theta}, \quad W = \sqrt{U} e^{-i\theta}.$$

The angular integral vanishes unless  $a = b$ . In that case,

$$\mathbb{E}[Z^a W^a] = \mathbb{E}[U^a] = \int_0^\infty u^a e^{-u} du = a!.$$

$\square$

**Proposition 6.2** (Intertwining embedding). *The assignment*

$$\iota(Z) = z, \quad \iota(W) = uz^{-1}$$

*defines an injective algebra homomorphism  $\iota : \mathbb{C}[Z, W] \hookrightarrow \mathcal{A}_1$  satisfying*

$$\Gamma_1(\iota(P)) = \mathbb{E}[P(Z, W)] \quad (P \in \mathbb{C}[Z, W]). \quad (6.2)$$

*Moreover, the torus-weight support of  $\iota(P)$  is exactly the angular weight support*

$$\Omega_{\text{ang}}(P) = \{a - b : [Z^a W^b]P \neq 0\}.$$

*Proof.* For a monomial,

$$\iota(Z^a W^b) = u^b z^{a-b}.$$

The pairs  $(b, a - b)$  determine  $(a, b)$  uniquely, so  $\iota$  is injective. Furthermore,

$$\Gamma_1(u^b z^{a-b}) = b! \mathbf{1}_{a=b} = a! \mathbf{1}_{a=b} = \mathbb{E}[Z^a W^b]$$

by Lemma 6.1. Linearity proves (6.2), and the support statement is immediate.  $\square$

**Theorem 6.3** (Two-variable Gaussian classification). *Let  $P \in \mathbb{C}[X, Y]$ , viewed in circular coordinates as an element of  $\mathbb{C}[Z, W]$ . The following are equivalent.*

- (i)  $\mathbb{E}[P(X, Y)^m] = 0$  for all sufficiently large  $m$ ;
- (ii)  $\mathbb{E}[P(X, Y)^m] = 0$  for every  $m \geq 1$ ;
- (iii) every angular weight  $a - b$  occurring in  $P$  is strictly positive, or every such weight is strictly negative.

Moreover, whenever these conditions hold, for every  $Q \in \mathbb{C}[X, Y]$ ,

$$\mathbb{E}[Q(X, Y)P(X, Y)^m] = 0 \quad (m \gg 0).$$

If the angular support of a nonzero  $P$  contains zero in its convex hull, then there is a  $k \geq 1$  such that

$$\mathbb{E}[P(X, Y)^{kp}] \neq 0$$

for every sufficiently large prime  $p$ .

*Proof.* Apply Theorems 1.1 and 1.2 to  $\iota(P)$  and use Proposition 6.2. Since the angular weights lie in the one-dimensional lattice  $\mathbb{Z}$ , exclusion of zero from their convex hull is equivalent to all occurring weights having one strict sign.  $\square$

**Corollary 6.4** (Gaussian Moments Conjecture in dimension two). *The Gaussian Moments Conjecture GMC(2) is true.*

*Proof.* If  $\mathbb{E}[P^m] = 0$  for every  $m \geq 1$ , the mixed-moment conclusion in Theorem 6.3 gives

$$\mathbb{E}[QP^m] = 0 \quad (m \gg 0)$$

for every polynomial  $Q$ .  $\square$

We finish by recording the orthogonal nullcone interpretation. In circular coordinates, consider the one-parameter subgroup

$$(Z, W) \mapsto (tZ, t^{-1}W). \tag{6.3}$$

In the original coordinates it is

$$\lambda(t) = \frac{1}{2} \begin{pmatrix} t + t^{-1} & i(t - t^{-1}) \\ -i(t - t^{-1}) & t + t^{-1} \end{pmatrix}. \tag{6.4}$$

It preserves  $X^2 + Y^2 = 2ZW$  and has determinant one, so  $\lambda(t) \in \mathrm{SO}_2(\mathbb{C})$ .

**Corollary 6.5** (Orthogonal nullcone in dimension two). *Let  $P \in \mathbb{C}[X, Y]$  satisfy  $\mathbb{E}[P(X, Y)^m] = 0$  for every  $m \geq 1$ . Then there is a one-parameter subgroup  $\lambda : \mathbb{C}^\times \rightarrow \mathrm{SO}_2(\mathbb{C})$  such that*

$$P(\lambda(t)(X, Y)) \in t\mathbb{C}[t][X, Y].$$

*Thus the nullcone conjecture of Derksen, van den Essen, and Zhao holds in dimension two.*

*Proof.* By Theorem 6.3, the angular weights of  $P$  are all positive or all negative. In the positive case, substitution (6.3) multiplies  $Z^a W^b$  by  $t^{a-b}$ , and every exponent is at least one. In the negative case, use the inverse one-parameter subgroup. The zero polynomial is immediate.  $\square$

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